

Logarithmic equations



1

a $y = 100$

b $y = 16\,807$

c $y = 343$

d $y = 36$

e $y = \sqrt[6]{6}$

f $y = 4$

2

a $x = 2$

b $x = 125$

c $x = \frac{64}{5}$

d $x = 100$

e $x = 7^{\frac{5}{4}}$

f $x = 15$

g $x = 6$

h $x = \frac{6556}{7}$

i $x = 6 - \sqrt{3}$

j $x = 32$

k $x = 4$

l $x = 2$

m $x = 137$

n $x = 4$

3

a $x = 5$

b $x = \frac{125}{4}$

c $x = 8$

d $x = 1406$

e $x = \frac{7}{2}$

f $x = \frac{6}{11}$

g $x = \frac{3}{13}$

h $x = 4$

i $x = -\frac{15}{7}$

j $x = 4$

k $x = \frac{343}{6}$

l $x = 45$

m $x = \frac{5}{16}$

n $x = \frac{131}{12}$

o $x = \frac{81}{5}$

4

a $x = 3$

b $x = 1, 6$

c $x = 2$

d $x = 20$

e $x = 64$

f $x = 5$

g $x = 2$